ArtMatch | Files and Experiments

Submission description | Team RB: Aesha Gadhiya & Lizi Maisuradze | OTH Amberg–Weiden

Project. ArtMatch retrieves five visually related artworks from a curated collection of 189 CU campus artworks across four floors. It returns artwork IDs, floor locations, similarity scores, and estimated visual attributes. The intended input is an **artwork image**, not an arbitrary photograph.

1 Data and preparation

CU collection (application gallery). Artwork regions were cropped from source photographs to reduce influence from walls, frames and background. Images were organized by floor and assigned stable IDs (CU_001–CU_189). Metadata records the ID, filename and floor. The CU collection supplies a retrieval gallery, **not similarity ground-truth labels**.

WikiArt (training and validation). A 3,000-image subset spanning Impressionism, Realism, Expressionism, Cubism and Abstract Expressionism was selected. Style labels define positive and negative pairs for supervised contrastive learning. The artist-disjoint split has 2,500 training and 500 validation paintings (500/100 per style). Split audits detected no artist overlap and no exact or near-duplicate overlap under the implemented checks.

Test split. No independent test partition was created. Validation was used to select a checkpoint and compare methods, so reported results are **validation results, not independent test performance**. A held-out test split and human-rated CU relevance judgments remain future evaluation work.

2 Data files and what they do

File / folder	Purpose
processed_gallery/images/floor_1 ... floor_4/	Cropped CU artworks, 67 / 65 / 38 / 19 images by floor (189 total).
processed_gallery/gallery_metadata.csv	Maps CU artwork IDs to filenames and floor locations.
embeddings/cu_clip_baseline.npz	Cached L2-normalized, 512-dimensional frozen CLIP features for the 189 CU artworks.
training_data/wikiart_style_splits.csv	Reproducible WikiArt subset and artist-disjoint train/validation assignments.
training_data/wikiart_clip_features.npz	Cached CLIP features, style labels and split indicators for 3,000 selected WikiArt paintings.
models/retrieval_projection_head.pt	Saved trained projection-head checkpoint (256-dimensional output; best validation epoch 2).

Packaging note: Depending on the submission archive, image files, feature caches and the checkpoint may be provided separately or via documented access links. The code alone cannot run a complete live retrieval demo without the CU images and the trained checkpoint.

Source files and experiments

Implementation map and verified results

3 Source files

File	Contents
artmatch_baseline.ipynb	Loads pretrained CLIP ViT-B/32; preprocesses/caches CU gallery embeddings; performs original-CLIP top-five retrieval and attribute analysis.
artmatch_training.ipynb	Selects/splits WikiArt data, audits overlap, extracts frozen CLIP features, trains and evaluates the projection head, examines extended training, and prepares adapted CU retrieval.
artmatch_gallery_ui.py	Local browser application: artwork upload, ranked CU matches, IDs/floors, similarity scores and estimated visual descriptors.
requirements.txt	Python dependency list; not a dataset or model artifact.
README.md	Project summary, setup requirements, file roles and reproducibility limitations.

4 Experiments and findings

Experiment	Method and finding
1 Original CLIP baseline	Use pretrained, frozen CLIP image embeddings (512-D) and cosine similarity. On the WikiArt validation set, same-style Precision@5 = **0.7264**. No WikiArt-specific CLIP training.
2 Style-adapted retrieval	Train only a 512 → 256 → 256 MLP projection head on cached WikiArt CLIP embeddings using supervised contrastive loss (Adam, LR 0.0003, temperature 0.1, dropout 0.2). Best checkpoint: epoch 2; validation P@5 = **0.7456**.
3 Extended-training analysis	Repeat training for 30 epochs without early stopping. Training P@5 reaches **1.0000**, but validation P@5 falls to **0.6632**: evidence of overfitting. Original training stopped at epoch 7 and restored epoch 2.
Application demonstration	Apply baseline and adapted embeddings to CU artworks; compare top-five ranks for the same artwork query. Attribute descriptions combine pretrained CLIP image–text scoring and pixel-based color analysis.

5 Interpretation and limitations

The measured gain of **1.92 percentage points** concerns **same-style retrieval on WikiArt validation data**, not verified CU retrieval accuracy. CU examples demonstrate rankings, but no human-annotated CU relevance benchmark has been completed. Some generated attributes remain uncertain; color heuristics and conservative handling of abstract-art object labels were refined after qualitative inspection.

Experiment organization. The baseline and trained/extended experiments are documented in the two notebooks above; a separate notebook was not created for the 30-epoch analysis. The submission can group these materials under **Sources/Experiment_1**, **Experiment_2** and **Experiment_3** with explanatory notes, without claiming any additional runs.